Proof Verification Dossier

Bounds and Estimators for the Civilian-to-Combatant Death Ratio

Internal document for manual checking — not for submission.

Machine companions: `analysis/verify_proofs.py` (sympy + Monte Carlo, all checks pass) and `formal/casualty_proofs` (Lean 4 + mathlib, kernel-checked, no `sorry`s).

July 6, 2026

How to read this document. Each claim is stated exactly as in the manuscript, followed by a proof written in more detail than the paper's appendix, an explicit list of the assumptions used at each step, and a verification note describing the symbolic (sympy) or Monte-Carlo check performed. The algebraic claims (1–4, the M-monotonicity step of Claim 7, and the Section 6 arithmetic over exact rationals) are additionally machine-checked in Lean 4 with mathlib (`formal/casualty_proofs`); the Lean development compiles with no unproven statements. Three errors were found in the first verification round and corrected in the manuscript; a second round of independent external review of this dossier surfaced further defects (most materially in the former Proposition 5.1), also now corrected in both dossier and manuscript. All are flagged inline and catalogued in the final two sections.

Contents

1 Setup and standing conventions

Population of size N; deaths D in the window $[t_0, t_1]$. Classes: W (women of all ages + males under 18) with population share w, and AM (males 18+) with population share a. *Convention (revised in this version):* AM is *all* males 18+, so

$$w + a = 1 \quad \text{exactly.}$$

(The prior draft used a = males 18–60 = 0.255, leaving men over 60 unassigned; this left $w + a = 0.988$ and made $a = 1 - w$ substitutions silently wrong. Now $a = 0.267$.)

Combatant stock M, $f = M/N$, all combatants in AM (slack for exceptions handled in Claim 3). Estimand $q = D_M/D$, the combatant share of the dead; ω = share of the *dead* in class W.

The hazard model. All identification results use one structural model: conditional on being a civilian, a member of class W dies with per-capita hazard λ and a member of civilian-AM with hazard $\mu\lambda$, $\mu \geq 1$. Combatant deaths are governed separately (their total is qD by definition). No assumption is made about λ itself — it cancels.

2 Claim 1: benchmark identification (paper eq. 2, Assumption 3.1)

Claim 1 (Point identification under exchangeable collateral)**.** *If all civilians face equal per-capita hazard ($\mu = 1$), then*

$$q = 1 - \frac{\omega(1-f)}{w}.$$

Proof. Civilian population masses: class W has wN civilians (no combatants in W); civilian AM has $(a-f)N$. With equal hazard λ, expected civilian deaths in W and civilian-AM are λwN and $\lambda(a-f)N$, so the fraction of *civilian* deaths landing in W is

$$\frac{wN}{wN + (a-f)N} = \frac{w}{w + a - f} = \frac{w}{1-f},$$

where the last step uses $w + a = 1$. Combatant deaths (share q of all deaths) land entirely in AM. Hence the share of *all* deaths in W is

$$\omega = (1-q) \cdot \frac{w}{1-f},$$

and solving for q gives the display. Note this is exact in expectation (a first-moment identity), not an approximation; finite-sample noise is handled by the delta method (Claim 5). □

Verification status. **VERIFIED**. Sympy: solving $\omega = (1-q)w\lambda/(w\lambda + (a-f)\mu\lambda)$ for q and substituting $\mu = 1$, $a = 1 - w$ reproduces the display exactly (λ cancels symbolically). Degenerate cases: $\omega = w$, $f = 0 \Rightarrow q = 0$; $\omega = 0 \Rightarrow q = 1$. *Note the subtlety*: the formula requires $a = 1 - w$; with the old convention $a = 0.255 \neq 1 - w$ the two displayed forms of the theorem disagreed by ∼1%. This motivated the convention change above.

3 Claim 2: sharp identified set (paper Theorem 3.3)

Claim 2 (Sharp identified set under μ-bounded exposure)**.** *Suppose $f \leq a$ and the hazard model holds for some $\mu \in [\mu_{\text{lo}}, \mu_{\text{hi}}]$. Given (ω, w, a, f),*

$$q(\mu) = 1 - \omega\,\frac{w + \mu(a-f)}{w},$$

is monotone decreasing in μ, and the identified set for q is exactly $[\,q(\mu_{\text{hi}}),\, q(\mu_{\text{lo}})\,] \cap [0,1]$.

Proof. Step 1 (the curve). With hazards $(\lambda, \mu\lambda)$, expected civilian deaths are proportional to $\lambda wN + \mu\lambda(a-f)N$, so the civilian share landing in W is $w/(w+\mu(a-f))$ and

$$\omega = (1-q)\,\frac{w}{w+\mu(a-f)}. \tag{1}$$

Solving (1) for q gives $q(\mu)$.

Step 2 (monotonicity). $\partial q/\partial\mu = -\omega(a-f)/w \le 0$, with equality iff $a = f$ (no civilian adult men) or $\omega = 0$.

Step 3 (validity: every point of the interval with $q(\mu^*) \in [0,1]$ is attainable). Fix any $\mu^* \in [\mu_{\text{lo}}, \mu_{\text{hi}}]$ such that $q(\mu^*) \in [0,1]$ (values outside $[0,1]$ are not shares and are excluded by the final intersection), and any $\lambda > 0$ small enough that all implied class death counts are below class populations. The pair $(\lambda, \mu^*\lambda)$ is an admissible hazard configuration whose implied ω satisfies (1) with $q = q(\mu^*)$. Hence $q(\mu^*)$ is consistent with the data (ω, w, a, f): the identified set contains $\{q(\mu) : \mu \in [\mu_{\text{lo}}, \mu_{\text{hi}}]\} \cap [0,1]$, which by continuity and monotonicity is $[q(\mu_{\text{hi}}), q(\mu_{\text{lo}})] \cap [0,1]$.

Step 4 (sharpness: nothing outside is attainable). Conversely, any admissible configuration has some $\mu \in [\mu_{\text{lo}}, \mu_{\text{hi}}]$, and (1) then forces $q = q(\mu)$, which lies in the interval. The intersection with $[0,1]$ imposes the logical range of a share. □

Verification status. **VERIFIED**. Sympy confirms the inversion and the derivative. 2,000 random parameter draws confirm endpoint ordering. The forward model's three death-shares (combatant, civilian-AM, W) sum to 1 symbolically. *Caveat worth knowing when checking manually:* the theorem treats ω as an exact population quantity; the "identified set" is a set for the estimand given error-free inputs. Sampling error enters separately via Claim 5 and systematic error via the anchor sensitivity analysis. This is stated in the paper's Discussion (limitation 3).

4 Claim 3: combatants outside AM (paper Remark 3)

Claim 3 (η-slack)**.** *If a fraction $\eta \in [0, \bar\eta]$, $\bar\eta = 0.03$, of combatant deaths falls in class W, then with $S := w/(w+\mu(a-f))$ the observable identity becomes $\omega = (1-q)S + \eta q$, so*

$$q_{\text{true}} = \frac{S-\omega}{S-\eta}, \qquad q_{\text{true}} - q_{\eta=0} \;=\; \frac{\eta(S-\omega)}{S(S-\eta)} \;=\; \frac{\eta\, q_{\text{true}}}{S} \;=\; \frac{\eta\, q_{\eta=0}}{S-\eta} \;\ge\; 0.$$

Ignoring the slack understates *q, by at most $\bar\eta q_{\eta=0}/(S-\bar\eta) \le 1.4$ percentage points at the Gaza values ($q_{\eta=0} \le 0.26$ and $S \ge 0.597$ for $\mu \le 2$; at $\mu = 1$, $S = 0.748$ and the bias is ≤ 1.1 points).*

Proof. Deaths in W now come from two sources: civilian deaths landing in W, which is $(1-q)S$ of all deaths as in Claim 2, plus the fraction η of combatant deaths, contributing ηq. Setting $\omega = (1-q)S + \eta q = S - q(S-\eta)$ and solving for q gives $q_{\text{true}} = (S-\omega)/(S-\eta)$. Subtracting $q_{\eta=0} = (S-\omega)/S$,

$$q_{\text{true}} - q_{\eta=0} = (S-\omega)\Big(\frac{1}{S-\eta} - \frac{1}{S}\Big) = \frac{\eta(S-\omega)}{S(S-\eta)},$$

which is nonnegative because $\omega \le S$ whenever $q \ge 0$. Substituting $S - \omega = q_{\text{true}}(S-\eta)$ gives the second form $\eta q_{\text{true}}/S$; substituting $S - \omega = q_{\eta=0}S$ gives the third form $\eta q_{\eta=0}/(S-\eta)$. Note the distinction: the bias equals $\eta q/(S-\eta)$ with q the *no-slack* estimate, and $\eta q/S$ with q the *true* share — the two coincide only to first order in η. □

Verification status. **ERROR FOUND & FIXED** (twice). The manuscript originally asserted the shift is "$-\bar{\eta}$" (i.e. subtract η from q). Symbolic verification shows this is wrong in *both* magnitude and sign: the true bias is $+\eta q_{\eta=0}/(S-\eta)$ (understatement, not overstatement, of q), and its magnitude is $\approx$0.011 at Gaza values ($\mu = 1$), not $\eta = 0.03$. A second error was caught by external review of a previous version of *this dossier*: the bias chain contained a spurious factor ($\eta q_{\text{true}}/S \cdot S/(S-\eta)$, which double-counts the $(S-\eta)^{-1}$ correction), and the claimed support "$S \geq 0.70$" fails for $\mu > 1.15$ at Gaza demographics ($S(1.5) = 0.664$, $S(2) = 0.597$). Both are corrected above; the bias bound is now stated per-μ. Sympy check: all three displayed forms are now symbolically equal to $q_{\text{true}} - q_{\eta=0}$. Direction note unchanged: the slack makes the paper's *upper* bounds conservative in the claim-favourable direction, so no downstream conclusion changes.

5 Claim 4: manpower bound (paper Corollary 3.4)

Claim 4. *$q \leq M/D$; whenever $\max\{0, q(\mu_{\text{hi}})\} \leq \min\{1,\ q(\mu_{\text{lo}}),\ M/D\}$, the joint identified set is $\big[\max\{0, q(\mu_{\text{hi}})\},\ \min\{1,\ q(\mu_{\text{lo}}),\ M/D\}\big]$; otherwise it is empty and the inputs are jointly infeasible.*

Proof. Combatants killed cannot exceed combatants existing: $D_M \leq M$. Divide by $D > 0$. The joint set is the intersection of two intervals, hence the endpoint formulas. (If the intersection is empty, the inputs are jointly infeasible — this is exactly the event detected by a positive contradiction radius.) □

Verification status. **VERIFIED** (algebraic identity). One caveat for manual review: M here is the stock at t_0; if recruitment during the war is material, M should be the stock-plus-inflow, which *raises* the bound. In the Gaza application the bound is slack by a factor > 2 ($M/D = 0.64$ vs. $q \leq 0.26$), so recruitment cannot change any conclusion.

6 Claim 5: delta-method standard error (paper §3)

Claim 5. *For $\hat{q} = 1 - \hat{\omega}(1-\hat{f})/\hat{w}$ with independent inputs,*

$$\widehat{\text{Var}}(\hat{q}) \approx \frac{\hat{\omega}^2}{\hat{w}^2}\widehat{\text{Var}}(\hat{f}) + \frac{(1-\hat{f})^2}{\hat{w}^2}\widehat{\text{Var}}(\hat{\omega}) + \frac{\hat{\omega}^2(1-\hat{f})^2}{\hat{w}^4}\widehat{\text{Var}}(\hat{w}).$$

Proof. First-order Taylor expansion of $g(\omega, f, w) = 1 - \omega(1-f)/w$:

$$\partial_f g = \frac{\omega}{w}, \qquad \partial_\omega g = -\frac{1-f}{w}, \qquad \partial_w g = \frac{\omega(1-f)}{w^2}.$$

$\text{Var}(\hat{q}) \approx \sum(\partial g)^2\text{Var}$, using independence (no covariance terms). Squaring the gradients gives the display. □

Verification status. **VERIFIED**. All three partial derivatives and the assembled formula verified symbolically. Assumption to flag for manual review: independence of $(\hat{\omega}, \hat{f}, \hat{w})$ is plausible here because they come from different institutions (deaths records, military-balance estimates, census), but any common-mode error (e.g. both ω and w derived from the same census frame) would add covariance terms. The paper's conservative $\sigma_\omega = 0.01$ (5× the binomial SE) partially absorbs this.

7 Claim 6: precision-weighted aggregation (paper Proposition 2.1)

Claim 6. *For independent unbiased sources with variances σ_i^2 treated as known (or fixed externally; with estimated $\hat{\sigma}_i^2$ the optimality is conditional on the variance estimates, exact asymptotically), the estimator*

$$\hat{\theta}_{\text{agg}} = \frac{\sum_i \hat{\theta}_i/\hat{\sigma}_i^2}{\sum_i 1/\hat{\sigma}_i^2}$$

is minimum-variance among linear unbiased combinations with weights fixed given the variances. With per-source biases b_i, the aggregate bias is the precision-weighted mean of the b_i. Under ε_i-contamination in which the contaminated report is displaced from the truth by at most R_i (equivalently, the contaminating support is contained in a set of diameter R_i containing the truth), the worst-case aggregate bias is at most $\sum_i \varepsilon_i R_i\, \hat{\sigma}_{\text{agg}}^2/\hat{\sigma}_i^2$.

Proof. (i) Minimise $\sum c_i^2 \hat{\sigma}_i^2$ subject to $\sum c_i = 1$. The Lagrangian condition gives $2c_i\hat{\sigma}_i^2 = \kappa$, i.e. $c_i \propto 1/\hat{\sigma}_i^2$; the constraint fixes the normalisation. The objective is strictly convex, so this is the unique minimum (Gauss–Markov in one dimension).

(ii) $\mathbb{E}\hat{\theta}_{\text{agg}} - \theta = \sum_i c_i b_i$ with $c_i = \hat{\sigma}_{\text{agg}}^2/\hat{\sigma}_i^2$; direct substitution.

(iii) Contamination replaces source i's report with probability ε_i by a value displaced by at most R_i (diameter of the contaminating support). The expected displacement of source i's contribution is at most $\varepsilon_i R_i$, weighted into the aggregate by c_i; summing over i gives the bound. □

Verification status. **VERIFIED**. Optimality confirmed on 300 random configurations against random competitor weights; bias formula confirmed symbolically. Nuance for manual review: in (iii) "diameter of the support" means the contaminated value cannot leave a set of diameter R_i around truth; if the adversary is unbounded, so is the worst case — this is why the paper pairs the proposition with the identified-set diameters R_i (removal diameters), which are finite by construction.

8 Claim 7: feasibility and contradiction (paper Theorem 4.3)

Claim 7. *Fix $(\hat{a}, \hat{f}, \hat{D})$ and $[\mu_{\text{lo}}, \mu_{\text{hi}}]$. Define $\mathcal{F}(x)$ as in the paper (eq. 4) and*

$$\rho^{(i)} = \inf_{\omega', w', M'} \max\Big\{ \tfrac{|\omega'-\hat{\omega}|}{\sigma_\omega}, \tfrac{|w'-\hat{w}|}{\sigma_w}, \tfrac{(M'-\hat{M})_+}{\sigma_M} \Big\} \ \text{s.t. } q^{(i)} \in \mathcal{F}((\omega', w', \hat{a}, \hat{f}, M', \hat{D})).$$

Then (i) $\rho^{(i)} = 0$ iff $q^{(i)} \in \mathcal{F}(\hat{x})$, and (ii) if $\rho^{(i)} > 0$, every rationalisation (ω', w', M') has at least one of $|\omega' - \hat{\omega}|/\sigma_\omega$, $|w' - \hat{w}|/\sigma_w$, $(M' - \hat{M})_+/\sigma_M$ at least $\rho^{(i)}$; that is, at least one input must move by $\rho^{(i)}$ standard errors in a penalised direction *(downward moves of M are free by construction, since lowering the manpower ceiling can only hurt the claim).*

Proof. (i, ⇐) If $q^{(i)} \in \mathcal{F}(\hat{x})$, take $(\omega', w', M') = (\hat{\omega}, \hat{w}, \hat{M})$: objective 0.

(i, ⇒) Suppose $\rho^{(i)} = 0$. Take a sequence (ω'_n, w'_n, M'_n) of feasible points with objective $\to 0$. Because the M-penalty is *one-sided*, this does *not* imply $M'_n \to \hat{M}$; it implies only

$$\omega'_n \to \hat{\omega}, \qquad w'_n \to \hat{w}, \qquad (M'_n - \hat{M})_+ \to 0,$$

i.e. $\limsup_n M'_n \le \hat{M}$ (with M'_n possibly far below $\hat{M}$). We therefore argue via the constraint rather than via convergence of M'_n. Each feasible point satisfies (a) $q^{(i)} = 1 - \omega'_n(w'_n + \mu_n(a - f))/w'_n$ for some $\mu_n \in [\mu_{\text{lo}}, \mu_{\text{hi}}]$, and (b) $q^{(i)} D \le M'_n$. For (a): pass to a subsequence with $\mu_n \to \mu^*$ in the compact interval; continuity of $(\omega', w', \mu) \mapsto 1 - \omega'(w' + \mu(a - f))/w'$ (on

$w' > 0$) gives $q^{(i)} = 1 - \hat{\omega}(\hat{w} + \mu^*(a - f))/\hat{w}$, so the exposure constraint holds at the observed $(\hat{\omega}, \hat{w})$. For (b): $q^{(i)} D \leq M'_n \leq \hat{M} + (M'_n - \hat{M})_+ \to \hat{M}$, so $q^{(i)} D \leq \hat{M}$. Hence $q^{(i)} \in \mathcal{F}(\hat{x})$. (Note the repaired step: the naive claim "objective $\to 0$ implies the full vector converges to $(\hat{\omega}, \hat{w}, \hat{M})$" is false under the one-sided penalty; only the inequality $q^{(i)} D \leq \hat{M}$ survives the limit, and that is all the theorem needs.)

(ii) Let (ω', w', M') be any point of C ("rationalisation"). By definition of the infimum, $\max\{|\omega' - \hat{\omega}|/\sigma_\omega,\ |w' - \hat{w}|/\sigma_w,\ (M' - \hat{M})_+/\sigma_M\} \geq \rho^{(i)}$, so at least one of the three *penalised* displacements is $\geq \rho^{(i)}$ SEs. This does not assert "M moved" generically: lowering M' costs nothing under the objective (and only shrinks the feasible set), so a positive radius can only ever be certified by an upward move of M, or a move of ω or w in either direction. □

Verification status. **ERROR FOUND & FIXED** (proof only; statement repaired in wording). External review of a previous version of this dossier correctly observed that the original proof of (i,⇒) claimed $(\omega'_n, w'_n, M'_n) \to (\hat{\omega}, \hat{w}, \hat{M})$, which is false under the one-sided M-penalty (downward moves of M' are free). The repaired proof above passes to the limit through the constraint $q^{(i)} D \leq M'_n \leq \hat{M} + o(1)$ instead, and part (ii) now states explicitly which displacements are penalised. The theorem's *conclusion* is unchanged. Numerically, 2,000 random infeasible claims all fail to become feasible under 10^{-9}-perturbations (no boundary pathologies found). The reduction used in the code (radius on the ω-axis = distance from $\hat{\omega}$ to the interval $[\omega_{\text{needed}}(\mu_{\text{hi}}), \omega_{\text{needed}}(\mu_{\text{lo}})]$) is justified by ω_{needed} decreasing in μ (verified symbolically). *Related error found and fixed elsewhere*: the original *application* of this definition in Section 6 evaluated the radius at the least favourable μ instead of minimising over the admissible range, overstating ρ ($\gtrsim 20$ claimed; correct values 8–31 depending on anchor and exposure regime, and $\rho = 0$ for the 17k endpoint under agnostic exposure on the MoH anchor). The definition and theorem themselves were always correct.

9 Claim 8: bias-robust sensitivity band (paper Proposition 5.1, restated)

Claim 8. *Let $I_0 = [L_0, U_0]$ be the $1 - \alpha$ credible interval under face-value sources, posterior supported on the identified set implied by the sources used. Each source i is independently contaminated with probability ε_i; removing the sources in T leaves an identified set of diameter R_T ($R_i := R_{\{i\}}$). Then: (i) conditional on realised pattern T, every attainable $1 - \alpha$ credible interval lies within $[L_0 - R_T,\ U_0 + R_T]$; (ii) the* expected *endpoint displacement is at most $\sum_i \varepsilon_i R_i$ plus $O(\varepsilon^2)$ terms involving multi-removal diameters, so $I_\varepsilon = [L_0 - \sum_i \varepsilon_i R_i,\ U_0 + \sum_i \varepsilon_i R_i]$ is a first-order sensitivity band,* not *a uniform envelope over realised intervals; (iii) if the posterior mixture weight on contaminated components is at most $\tau := 1 - \prod_i (1 - \varepsilon_i)$ and the face-value posterior density exceeds $m > 0$ on the segment between the clean and contaminated quantiles,*

$$|\hat{Q}_\varepsilon(u) - \hat{Q}_0(u)| \ \leq\ m^{-1}\tau, \qquad u \in \{\alpha/2,\ 1 - \alpha/2\}.$$

Proof. *(i)* Condition on T: honest sources outside T confine θ to the removal identified set of diameter R_T, which contains the face-value value; any posterior formed under adversarial reports from T is supported there, so each credible-interval endpoint is within R_T of the corresponding face-value endpoint.

(ii) The pattern probability is $P(T) = \prod_{i \in T} \varepsilon_i \prod_{i \notin T}(1 - \varepsilon_i)$, so the expected displacement is at most $\sum_T P(T) R_T = \sum_i \varepsilon_i R_i \prod_{j \neq i}(1 - \varepsilon_j) + \sum_{|T| \geq 2} P(T) R_T \leq \sum_i \varepsilon_i R_i + O(\varepsilon^2)$. This is a statement about the *mean* over the contamination pattern. A quantile (interval endpoint) of the realised posterior can move by up to R_T on a contamination event of probability ε_i: with one source, $\varepsilon = 0.04$, $R = 100$, a contaminated component at distance $\approx R$ carrying 4% posterior mass moves the 97.5% quantile by $\approx R$, far beyond $\varepsilon R = 4$. Hence the band

is first-order in ε, exact when at most one source can be contaminated *and* displacement is measured in expectation.

(iii) If the posterior mixture weight on contaminated components is $\le \tau$, the contaminated and face-value posterior CDFs differ by at most τ uniformly; the mean value theorem under the density floor m (on the whole segment between the two quantiles) gives the display. The weight condition does *not* follow from the prior probabilities alone: posterior component weights are $\propto P(T)\,Z_T$ where Z_T is the component marginal likelihood, so an adversarial component that fits the data better than the clean model carries posterior weight exceeding τ. □

Verification status. **ERROR FOUND & FIXED** (statement was materially over-claimed). External review of a previous version of this dossier exhibited the counterexample reproduced in (ii): the original statement asserted that I_ε *contains every* attainable credible interval, which is false for tail quantiles — the mixture argument controls an expectation, not an envelope. Numerically confirmed: with $\varepsilon = 0.04$, $R = 100$, clean posterior $\mathcal{N}(0, 0.1^2)$ and contaminated component $\mathcal{N}(100, 0.1^2)$, the pooled 97.5% quantile is 99.97 while $\varepsilon R = 4$. The same review identified the posterior-weight gap in the old part (ii): mixture weights after conditioning are $P(T)Z_T$-proportional, not $P(T)$. The proposition has been restated in the manuscript as a *patternwise envelope + first-order sensitivity band + quantile bound under an explicit posterior-weight condition*, and all prose referring to it as a "credible interval that contains every adversarial interval" has been reworded. Downstream use in Section 6 is unaffected in substance: the quoted 25.1% → 28.7% inflation is the first-order band I_ε, now labelled as such; the paper's headline rejections rest on the bounds and the contradiction radius, not on this proposition.

10 Section 6 arithmetic (spot summary)

All application numbers are generated by `analysis/validate_bounds.py`. Key identities re-verified here at $(w, a, f, \omega) = (0.733, 0.267, 0.020, 0.560)$:

Quantity	Value	Check
$q(1)$	25.13%	symbolic + numeric
$q(1.5)$	15.69%	numeric (prior table misprinted 15.72%)
$q(2)$	6.26%	numeric
root of $q(\mu) = 0$	$\mu^* = 2.33$	numeric; matches the q=0 symmetry discussion
μ needed for $q = 17/70$	1.04	inversion; ≥ 1 so arithmetically feasible
μ needed for $q = 25/70$	0.44	inversion; < 1 so infeasible for any admissible μ
M/D	0.643	non-binding vs. all bounds above

11 Errors found by this verification (all fixed in the manuscript)

1. **Remark 3 (η-slack) was wrong in sign and magnitude.** Claimed shift "$-\bar{\eta}$"; correct bias is $+\eta q/(S - \eta) \approx 1$ percentage point at Gaza values (ignoring the slack *understates* q). Rewritten with the exact expression. Downstream conclusions unaffected (direction is claim-favourable).

2. **The contradiction radius was originally computed at the wrong μ.** Definition 4.2 grants the claim the most favourable μ in the admissible range; the original Section 6 evaluated at the least favourable, producing "$\rho \gtrsim 20$". Corrected values: $\rho_\omega \approx 7.9$–31 depending on anchor and exposure regime, and $\rho = 0$ for the 17k endpoint under agnostic exposure on the MoH anchor (it is rejected by exposure calibration, not arithmetic). Disclosed in the response letter.

3. **Demographic classes did not partition the population.** $a = 0.255$ (males 18–60) left men over 60 unassigned while proofs used $a = 1 - w$. Convention changed to AM = males 18+, $a = 0.267$; all numbers regenerated.

12 Errors found by external review of this dossier (second round, all fixed)

A previous version of this dossier was independently reviewed; the following defects in the *dossier and manuscript statements* (as distinct from the application numbers) were confirmed and corrected:

1. **Claim 3's displayed bias chain contained a spurious factor.** The chain $\eta q_{\text{true}}/S \cdot S/(S-\eta)$ double-counts the $(S-\eta)^{-1}$ correction; the correct equalities are $\eta(S-\omega)/(S(S-\eta)) = \eta q_{\text{true}}/S = \eta q_{\eta=0}/(S-\eta)$. Verified symbolically; fixed here and in manuscript Remark 3, which now names which q appears in which form.

2. **Claim 3's numerical support "$S \geq 0.70$" was false for $\mu > 1.15$.** At Gaza demographics $S(1.5) = 0.664$, $S(2) = 0.597$. The bias bound is now stated per-μ (max 1.4 points at $\mu = 2$).

3. **Claim 7's proof mishandled the one-sided M-penalty.** "Objective $\to 0$ implies the full vector converges" is false when downward moves of M are free. Repaired by passing to the limit through the constraint $q^{(i)}D \leq M_n' \leq \hat{M} + o(1)$; theorem conclusion unchanged. Part (ii) now names the penalised displacements explicitly. Manuscript Theorem 4.3 statement and appendix proof updated.

4. **Claim 8 (Proposition 5.1) was materially over-claimed as a credible-interval envelope.** A one-source counterexample ($\varepsilon = 0.04$, $R = 100$) moves the 97.5% quantile by $\approx 100 \gg \varepsilon R = 4$. The mixture argument controls an *expectation*; additionally, posterior mixture weights are $P(T)Z_T$-proportional, not $P(T)$. Restated in the manuscript as patternwise envelope + first-order sensitivity band + quantile bound under an explicit posterior-weight condition; all "contains every adversarial interval" prose removed. The quoted Gaza inflation 25.1% $\to$ 28.7% is unchanged as a first-order band; no headline conclusion rests on this proposition.

5. **Smaller repairs.** Claim 2 Step 3 now restricts attainability to $q(\mu) \in [0, 1]$; Claim 4 states the nonempty-intersection condition explicitly; Claim 6 now says variances are treated as known and that R_i bounds displacement *from the truth*; the arithmetic table's $q(1.5)$ was misprinted as 15.72% (correct: 15.69%; the manuscript's rounded 15.7% was already correct).